

Customer Support Q&A; API

Node.js + OpenAI backend for lightweight support automation

Type Portfolio demo	Role AI Automation Developer	Stack Node.js, Express, OpenAI API, dotenv
-------------------------------	--	--

Problem

Many small businesses want AI-assisted customer support, but they do not need a full chatbot platform. They need a focused backend endpoint that can answer common questions using approved business information.

Solution

I built a small REST API that accepts a customer question, validates the input, sends it to OpenAI with a structured support context, and returns a concise customer-facing answer. The implementation includes environment-based API keys, timeout handling, rate-limit mapping, and setup documentation.

API Workflow

1. Client Request POST /support/ask with JSON question	2. Validation Check required field, empty text, and length	3. OpenAI Use support context and system prompt	4. JSON Response Return answer or mapped error
---	--	--	---

Key Features

- POST /support/ask endpoint for customer questions
- Support-focused system prompt that avoids inventing prices, policies, or guarantees
- Business context file that can be replaced with a client FAQ or knowledge base
- Environment-based OpenAI API key handling with .env
- Input validation for missing, empty, and oversized questions
- Rate limit, timeout, and provider error handling
- Health check endpoint for deployment checks
- README and small test suite for core logic

Implementation Snapshot

```
app.post("/support/ask", async (req, res) => {
  const validation = validateQuestionPayload(req.body);

  if (!validation.ok) {
    return res.status(400).json({
      error: "invalid_request",
      message: validation.message
    });
  }

  const result = await answerSupportQuestion({
    client,
    question: validation.question,
    model,
    timeoutMs
  });

  return res.json(result);
});
```

Business Value

- Creates a practical backend foundation for a customer support assistant
- Helps businesses answer common questions using approved support information
- Can power website chat widgets, contact forms, internal support tools, and FAQ assistants
- Keeps the first version small, understandable, and easy to extend

Portfolio Note

This is a portfolio demo, not a live client deployment. In a real project, the sample support context would be replaced with the client's approved FAQs, policies, services, escalation rules, and brand tone.